INCH-POUND

MIL-L-3976D
29 December 1989
SUPERSEDING
MIL-L-3976C
15 September 1982

MILITARY SPECIFICATION

LIGHTS, MARKER, CLEARANCE (SERVICE AND BLACKOUT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers amber (yellow) or red clearance marker lights for service and blackout use in military vehicles. The clearance marker light assembly shall be referred to as "light assembly" herein (see 6.1).

1.2 Classification. Lights shall be of the following classes and colors (see 6.2).

Class 1	- Service lights enable safe operations of military vehicles under normal conditions of visibility.
Class 2	- Blackout lights are designed to maintain vehicle blackout security during blackout conditions by limiting horizontal and vertical illumination as well as providing ample light for safe vehicle operation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6220

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1.2.2 Colors. The color of filters shall be designated as follows:

A	- Amber (Yellow).
B	- Red.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-349	- Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate.
O-O-670	- Orthophosphoric (Phosphoric) Acid, Technical.
W-L-111	- Lamps Incandescent (Electric, Miniature and Subminiature, Tungsten-Filament).
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-S-571	- Solder, Tin Alloy, Tin-Lead Alloy and Lead Alloy.
QQ-S-698	- Steel, Sheet and Strip, Low Carbon.
TT-E-527	- Enamel, Alkyd, Lusterless, Low VOC Content.
TT-P-664	- Primer Coating, Synthetic, Rust Inhibiting, Lacquer Resisting.
TT-S-1732	- Sealing Compound, Pipe Joint and Thread, Lead Free General Purpose.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-3976/1(AT)	- Light Emitting Diode (LED), Marker, Clearance Blackout, 14 and 28 Volt.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low Tension, Heavy Duty Single-Conductor and Multiconductor.

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MIL-T-13513	- Terminal, Lug, Splice, Conductor (Electrical Solderless, for Automotive Use).
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-C-81562	- Coatings, Cadmium, Tin Cadmium, and Zinc (Mechanically Deposited).

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Markings for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MS35420	- Lens, Clearance Marker Light, Blackout.
MS35421	- Lens, Clearance Marker Light, Service.
MS35422	- Lampholder, Light, Marker, Clearance.
MS35423	- Light, Marker, Clearance-Service.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A167	- Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
ASTM A176	- Standard Specification for Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip.
ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM D395	- Rubber Property-Compression Set, Standard Test Methods for.
ASTM D471	- Rubber Property-Effect of Liquids, Standard Test Method for.
ASTM D573	- Rubber-Deterioration in an Air Oven, Standard Test Method for.
ASTM D1149	- Rubber Deterioration-Surface Ozone Cracking in a Chamber, Standard Test Method for.
ASTM D2137	- Rubber and Rubber-Coated Fabrics-Brittleness Temperature by Impact, Standard Test Methods for.
ASTM D3935	- Standard Specification for Polycarbonate (Pc) Unfilled and Reinforced Material.
ASTM D3951	- Standard Practice for Commercial Packaging.
ASTM F104	- Standard Classification for Non-Metallic Gasket Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS STANDARDS

SAE J575	- Tests for Motor Vehicle Lighting Devices and Components.
SAE J578	- Color Specification.
SAE J592	- Clearance, Side Marker, and Identification Lamps.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a first article sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings (see 4.8.1).

3.2.1 Solder. Unless otherwise specified (see 6.2), solder shall be of a composition conforming to Sn 50, type RA or Sn 60, type RA of QQ-S-571 (see 4.8.1).

3.2.2 Lampholder boot rubber. Physical properties of the rubber used in the lampholder boot shall be as specified below and in table I (see 4.8.1):

a. Shore A durometer hardness: 50 ± 5
b. Minimum tensile strength: 1500 pounds per square inch (psi)
c. Minimum ultimate elongation: 3.5 percent (%)
d. Resistance to oil aging: 0-120% volume change

TABLE I. Material property values.

Physical property	Test
Resistance to heat aging: 70 hours, 158°F 1/	ASTM D573
Compression set: 50%, after 22 hours at 158°F	ASTM D395, method B
Resistance to Ozone at 50 parts per hundred million (pphm) at 100°F for 7 days	ASTM D1149
Low temperature brittleness at -67°F	ASTM 2137, method A
Resistance to water	ASTM D471

1/ Degrees Fahrenheit.

3.2.2.1 Lampholder door. Lampholder door shall be die cast aluminum as specified in 3.2.3 (see 4.8.1).

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3.2.2.2 Lampholder plate. Lampholder plate shall be die cast aluminum as specified in 3.2.3 (see 4.8.1).

3.2.3 Aluminum. Die cast parts shall be of an aluminum alloy conforming to composition A413.0, C443.0, A380.0, 380.0, or 360.0 of QQ-A-591 (see 4.8.1).

3.2.4 Lens assemblies.

3.2.4.1 Blackout lens assembly.

3.2.4.1.1 Uncolored lens. Uncolored lens shall be clear and shall be of molded polycarbonate plastic conforming to ASTM D3935 designation PC110B30720E01F02. Lens shall be painted black as specified in MS35420 (see 4.8.1).

3.2.4.1.2 Shield. Shield shall be made of 0.020 cold-rolled steel conforming to QQ-S-698. Shield shall be primed with primer conforming to TT-P-664, and finished with a black lusterless enamel conforming to TT-E-527 (see 4.8.1).

3.2.4.1.3 Filter. Filter shall be of molded cellulose acetate butyrate conforming to grade H2 of L-P-349. Color of filter shall be in accordance with Tennessee - Eastman Co. number 18029 (red) or 14686 (amber), formula 205A (see 4.8.1).

3.2.4.2 Service lens. Service lens shall be of molded polycarbonate plastic conforming to ASTM D3935 designation PC110B30720E01F02 for general purpose use or PC130B30720E01F02 for ultra violet stabilized. Lens shall be as specified in MS35421 (see 4.8.1).

3.2.5 Lampholder socket. Socket shall be made of 0.020 thick, cold-rolled sheet steel conforming to ASTM A167 or ASTM A176 (see 4.8.1).

3.2.6 Mounting pad. Mounting pad (gasket) shall be made of rubber-impregnated fiber, 0.062 inch thick, conforming to the identification number F337576M6 of ASTM F104 (see 4.8.1).

3.2.7 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.8.1 and 6.4).

3.2.8 Qualified products. When a part is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall maintain documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity (see 4.8.1).

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3.3 Design and construction. All parts shall be new and conform to the form and dimensions specified in MIL-L-3976/1(AT) and in MS35420, MS35421, MS35422, MS35423 and MS35424, as applicable (see 4.8.1 and 4.8.2).

3.3.1 Threaded parts. All screw threads shall conform to FED-STD-H28/2. The form, number per inch, and class of threads shall be as specified in applicable standards or drawings. All threads, except electrical conducting parts and self-locking nuts, shall be coated with antiseize compound conforming to TT-S-1732 (see 4.8.2).

3.3.2 Standard parts. Military standard parts shall be used wherever applicable. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.8.1 and 4.8.2).

3.3.3 Soldering. Soldering shall conform to requirement 5 of MIL-STD-454 (see 4.8.1 and 4.8.2).

3.3.4 Terminals. Terminals shall be soldered or crimped in accordance with requirement 5 of MIL-STD-454 or MIL-T-13513 (see 4.8.2).

3.3.5 Cable. Cable shall be no. 14 American Wire Gage (AWG) single conductor size conforming to type I, class A of MIL-C-13486 (see 4.8.1 and 4.8.2).

3.3.6 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with MIL-STD-889 (see 4.8.1 and 4.8.2).

3.3.7 Protective finishes, plating and sealing.

3.3.7.1 Lampholder socket. Socket shall be zinc plated to conform to type II, suffix Fe/Zn 25 of ASTM B633 or type II class 4 of MIL-C-81562 and then cleaned and the metal prepared with phosphoric acid conforming to O-O-670 prior to priming in accordance with TT-P-664 (see 4.8.2).

3.3.7.2 Aluminum parts. Aluminum parts shall be sealed before painting in accordance with MIL-C-5541 (see 4.8.2).

3.3.8 MS sheets. The individual requirements shall be as specified herein and in accordance with the applicable MS sheet. In the event of any conflict between the requirements of this specification and the MS sheet, the latter shall govern (see 4.8.1).

3.4 Performance.

3.4.1 Operation. The light shall operate without dimming or flickering when connected to a power source in accordance with the applicable drawing or MS (see 4.8.3.1).

3.4.2 Color. Color of light emitted shall be within the chromaticity coordinates of SAE J578 (see 4.8.3.2).

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3.4.3 Service light. With a 3-cp lamp installed, the light shall meet the minimum photometric requirements specified in SAE J592 (see 4.8.3.3).

3.4.4 Environmental. At the conclusion of each environmental category, the light assembly shall operate in accordance with the requirements of 3.4.1.

3.4.4.1 Temperature cycling. The light assembly shall show no evidence of cracking, deformation, opening of seals, or visual damage after exposure to extremes of high and low temperatures, and shall be resistant to the shock of alternate exposure to these extremes (see 4.8.4.1).

3.4.4.2 Vibration resistance. The light assembly shall show no evidence of disintegration after exposure to vibration testing under power for not less than 2 hours in each of 3 perpendicular axes for a total of 6 hours (see 4.8.4.2).

3.4.4.3 Shock resistance. The light assembly shall show no evidence of disintegration after induced shocks of sawtooth waveform of 6 minutes duration and a peak value of 75 gravity units (g) (see 4.8.4.3).

3.4.4.4 Waterproofness. The light assembly performance shall not be affected nor shall the light assembly show signs of leakage during and after submersion while under power (see 4.8.4.4).

3.5 Painting. Unless otherwise specified (see 6.2), the external surface of the light assembly shall be cleaned, primed, and finished in accordance with MIL-STD-193. The paint topcoat shall conform to MIL-C-46168 or MIL-C-53039. The color shall be lusterless camouflage forest green conforming to chip number 34083 of FED-STD-595 (see 4.8.2).

3.6 Marking. Marking shall be in accordance with MIL-STD-130 and shall consist of the MS part number and manufacturers identification (see 4.8.2).

3.7 Workmanship. Workmanship shall meet the requirements of MIL-STD-454, requirement 9 (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, it does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspections (QCI) (see 4.5).
c. Control tests (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 77 ± 18°F
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.4 First article inspection. When first article is required (see 3.1), first article inspection shall be performed on four first article samples (see 6.2). Unless otherwise specified (see 6.2), first article inspection shall include the inspections specified in table II (see 6.3.1 for first article inspection of spare parts).

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TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality Conformance Exam	Quality Conformance Tests	Control tests	Inspection of Packaging
Group A:							
Materials and construction	3.2 thru 3.2.8, 3.3.2 and 3.3.8	4.8.1	X				
Examination	3.3 thru 3.3.7 and 3.5 thru 3.7	4.8.2	X	X			
Group B: (Performance):							
Operation	3.4.1	4.8.3.1	X	X	X		
Color	3.4.2	4.8.3.2	X			X	
Service light	3.4.3	4.8.3.3	X			X	
Group C (Environment):							
Temperature cycling	3.4.4.1	4.8.4.1	X			X	
Vibration	3.4.4.2	4.8.4.2	X				
Shock resistance	3.4.4.3	4.8.4.3	X				
Waterproofness	3.4.4.4	4.8.4.4	X			X	
Group D (Packaging):							
Army	5.1	4.7.1	X				X
Navy/Air Force							
Level A	5.2	4.7.2	X				X
Level B	5.2	4.7.2	X				X
Commercial	5.2	4.7.2	X				X

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4.5 QCI. QCI shall include the examination of 4.5.1 and the tests of 4.5.2 (see 6.2). Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Examination. Light assemblies shall be examined for the defects specified in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	Faulty Grounded Power Lead Affecting Performance (see 3.3).	SIE 1/
Major		
101	Incorrect dimensions affecting interchangeability not within tolerance (see 3.3. 3.3.1, 3.3.2).	SIE
102	Operating inability (3.4.1).	Visual
103	Incorrect soldering (see 3.3.3 and 3.3.4).	Visual
104	Incorrect crimping of terminal (see 3.3.4).	Visual
105	Incorrect cable (see 3.3.5).	Visual
106	Incorrect dissimilar metals (see 3.3.6).	Visual
107	Faulty workmanship affecting performance (see 3.7).	Visual
Minor		
201	Incorrect dimensions not within tolerance but not affecting interchangeability (see 3.3).	SIE
202	Incorrect finish (see 3.5 and 3.3.7).	Visual
203	Incorrect marking (see 3.6).	Visual
204	Faulty workmanship not affecting performance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.2 QCI tests. Light assemblies shall be subjected to the QCI tests specified in table II.

4.6 Control tests. Light assemblies selected for control tests shall be new and subjected to the control tests specified in table II (see 6.2).

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4.7 Inspection of packaging.

4.7.1 Army. To determine conformance to 5.1, inspection of packaging shall be in accordance with the applicable packaging standard or packaging data sheet (see 5.1).

4.7.2 Navy and Air Force. To determine conformance to 5.2, the packaging shall be examined for the defects listed in table IV.

TABLE IV. Classification of defects (Packaging).

Category	Defect	Method of examination
Critical	None	
Major		
101	Improper level of packaging (see 5.2.1) or packing (see 5.2.2).	Visual
102	Parts improperly cleaned (see 5.2.1).	Visual
103	Unpainted surfaces not coated (see 5.2.1).	Visual
104	Light not properly cushioned (see 5.2.1).	Visual
105	Wrapping not of proper type, grade or class (see 5.2.1).	Visual
106	Incorrect unit carton (see 5.2.1).	Visual
107	Incorrect shipping container (see 5.2.2).	Visual
108	Illegible marking (see 5.2.3).	Visual
109	Incorrect marking (see 5.2.3).	Visual
Minor		
201	Improper closure and taping of package (see 5.1 and 5.2.1).	Visual

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.8, 3.3.2 and 3.3.8 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Examination. Conformance to 3.3 through 3.3.7 and 3.5 thru 3.7 shall be determined by examination for the defects listed in table III.

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4.8.3 Performance.

4.8.3.1 Operation. To determine conformance to 3.4.1, the assembled light shall have a 3-cp incandescent lamp installed in lampholder. The light shall be connected to the applicable power source (12, or 24 V dc) and observed.

4.8.3.2 Color. To determine conformance to 3.4.2, the transmitting media shall be tested in accordance with tristimulus test method specified in SAE J578 to determine conformance of red and amber (yellow) transmitting media.

4.8.3.3 Service light. To determine conformance to 3.4.3, the light shall be tested in accordance with the photometry test as specified in SAE J575.

4.8.4 Environment. At the conclusion of each environmental category, the light assembly shall be tested as specified in 4.8.3.1.

4.8.4.1 Temperature cycling. To determine conformance to 3.4.4.1, the light shall be tested as specified in method 107, test condition A of MIL-STD-202.

4.8.4.2 Vibration. To determine conformance to 3.4.4.2, after mounting to simulate actual mounting in use, and while operating, the light shall be subjected to the vibration resistance test specified in method 201 of MIL-STD-202 for one hour in each direction of the three major axes of the light. Filament breakage in lamp shall not constitute failure. During and after the test, the light shall operate as specified in 4.8.3.1.

4.8.4.3 Shock resistance. To determine conformance to 3.4.4.3, light, with the lamp installed, shall be mounted to simulate actual mounting in use and while operating, the light shall be subjected to the shock resistance test specified in method 213, test condition H (using the sawtooth waveform and 75 g peak value) of MIL-STD-202. Shocks shall be applied twice in each direction of the three major axes. Filament breakage in lamp shall not constitute failure. The light shall be observed and examined during and after the test.

4.8.4.4 Waterproofness. To determine conformance to 3.4.4.4, the light shall be tested in accordance with method 100, procedure 1 of MIL-STD-1184.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking (for Army use). Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

5.2 Preservation, packaging, packing, and marking (for Navy and Air Force use).

5.2.1 Preservation and packaging. Preservation and packaging shall be commercial or level A as specified (see 6.2).

5.2.1.1 Level A. Unless otherwise specified, each light shall be individually packaged in accordance with method IC-2 of MIL-P-116 in a unit container conforming to PPP-B-636, type CF, class domestic. Cushioning within the unit container shall be as required to prevent item damage. Box closure shall be in accordance with the appendix to the box specification. Level A unit packs shall be overpacked into an intermediate container, PPP-B-636, class NR, type CF. Quantities of unit packs per intermediate container shall be governed by the following limitations:

a. Maximum of 100 unit packs.
b. Maximum net load of 40 pounds.
c. Maximum size of 1.5 cubic feet with at least two dimensions not exceeding 16 inches.

5.2.1.2 Commercial. Lights shall be packaged in accordance with ASTM D3951.

5.2.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.2.1 Level A. Unless otherwise specified, a quantity of lights packaged as specified in 5.2.1 shall be packed in a wood, cleated plywood shipping container conforming to PPP-B-601, overseas type. Insofar as practicable, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

5.2.2.2 Level B. Level B shall be the same type as level A, except the shipping containers shall conform to domestic type.

5.2.2.3 Commercial. Lights shall be packed in accordance with the requirements of ASTM D3951.

5.2.3 Marking. In addition to any special markings required by the contract or order, interior containers and exterior containers shall be in accordance with the requirements of MIL-STD-129 (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Lights covered by this specification are primarily intended for mounting on military vehicles for use as combination clearance and sidemarker lights. Lights are either service or blackout class (see 1.2). Front and near front mounted lights must emit amber light. Rear and near rear mounted lights must emit red light.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class, color, and light to be furnished (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual document referenced (see 2.1.1 and 2.2).
d. If first article is required (see 3.1).
e. If soldering should be other than as specified (see 3.2.1).
f. If finish of exterior should be other than as specified (see 3.5).
g. If responsibility for inspection and the place of inspection is other than as specified (see 4.1).
h. If inspection conditions should be other than as specified (see 4.3).
i. Type of first article sample, and the number of samples to be inspected (see 4.4 and 6.3).
j. QCI acceptance criteria (see 4.5 and 6.7.1).
k. Control test sample size, frequency of tests, and disposition of lots covered by the sample size and of defective items (see 4.6 and 6.7.2).
l. For Army, selection of applicable level and packaging requirements (see 4.7.1 and 5.1).
m. For Navy and Air Force, specify level A or commercial preservation and packaging (see 4.7.2 and 5.2).
n. For Navy and Air Force, specify level A, B or commercial packing (see 4.7.2 and 5.2.2).
o. For Navy and Air Force, specify any special markings required (see 5.2.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisitions documents regarding arrangements for examinations, approval of first article test

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results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 First article inspection of spare parts. Spare parts, such as lenses (MS35420 and MS35421) and the lampholder (MS35422) should be considered as having met first article requirements, providing they have been installed in an applicable light assembly that has passed first article testing as specified in 4.4.

6.4 Definitions.

6.4.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.4.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Conditions of use of level B preservation. When level B preservation is specified (see 5.2.2), this level of protection should be reserved for the acquisition of light assemblies for resupply worldwide under known favorable handling, transportation, and storage conditions.

6.6 Subject term (key word) listing.

Blackout lights; for security.
Horizontal and vertical illumination.
Lights, clearance.
Service lights; safe operations of Military Vehicles.
Visibility.

6.7 Inspection. The following information is presented for guidance in specifying sampling and acceptance procedures for QCI, control testing, and packaging inspection in the acquisition document.

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6.7.1 QCI.

6.7.1.1 Sampling.

6.7.1.1.1 Lot formation. An inspection lot should consist of all the lights of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

6.7.1.1.2 Sampling for examination. Samples for quality conformance examination should be selected in accordance with general inspection level II of MIL-STD-105.

6.7.1.1.3 Sampling for test. Samples for tests should be selected in accordance with level S-3 of MIL-STD-105.

6.7.1.2 Examination.

6.7.1.2.1 Acceptable quality level. Each sample selected in accordance with 6.7.1.1.2 should be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

6.7.1.3 Test. Samples selected in accordance with 6.7.1.1.3 should be subjected to the quality conformance tests specified in table II, and should conform to an AQL of 6.5.

6.7.2 Control test. Control test should be conducted on samples selected at a rate of two (2) lights per month or two of each 500 lights produced, whichever occurs first, except that not more than four lights should be selected in any single month. The first sample should be taken at or near the start of production. Samples should be subjected to the applicable tests specified in table II.

6.7.2.1 Failure. Failure of a control test sample to pass any of the specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

6.7.3 Packaging inspection.

6.7.3.1 Sampling.

6.7.3.1.1 Lot formation. An inspection lot should consist of all the packs of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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6.7.3.1.2 Sampling for examination. Samples for examination should be selected in accordance with MIL-STD-105.

6.7.3.1.3 Sampling for test. One complete pack should be selected for acceptance testing.

6.7.3.2 Examination.

6.7.3.2.1 Acceptance quality level. Each sample selected in accordance with 4.7.1.2 should be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

6.7.3.2.2 Failure. Any package or pack containing one or more defects should be rejected, and if the number of defects in any one sample exceeds the acceptance number of that sample, the lot represented by that sample should be rejected.

6.7.3.3 Tests. The sample selected in accordance with 6.7.3.1.3 should be subjected to the applicable inspection provisions of MIL-STD-1186.

6.7.3.3.1 Failure. Failure of the pack to pass any specified test, may be cause for the Government to refuse to accept the lot until it has been proved, to the satisfaction of the Government, that the faults revealed by the tests have been corrected.

6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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